

Week 2 — Six questions for reading a Methods section

Duration: ~2.5 h (Tue, before the Thu lab)

Cap: **150 min**. Markers below total **122**; the remaining ~28 minutes are the room's — questions, the vote-and-argue moments, and the two live recomputations.

The lecture is one argument, not two halves. It opens with six questions a reader must be able to answer about any study, then fills them in with the demonstration paper, reaching for NHANES mechanics **at the point where a question cannot be answered without them**. The closing M0 section turns the six questions on the students' own candidate papers.

Row-set discipline. Every quantitative slide names the analysis set it is quoting. *Analysis set: defined here, then named on every slide that quotes one.* The reconstruction has two, and they are not interchangeable: **full reconstructed analytic file, N = 6,371** (what scripts 03–06 read) and **locked mortality-linked cohort, N = 6,048** (the WTSAF2YR subset). Saying “N” without saying which is the error this course exists to catch.

Learning objectives

1. Name the **research goal** a paper is pursuing — description, association, prognostic factor, prediction, or causal — and say what evidence that goal would require.
2. Answer the **six questions** for a published study, and show where the paper leaves one of them under-specified.
3. Locate any NHANES quantity a study uses: the **component file**, the **variable name**, and the **join key**.
4. Distinguish a **design feature** from **missingness**, and a **row set** from a cohort.
5. Say what a crude-to-adjusted change does and does not establish.

Continuity

- **Last week (Wk1):** reproducibility as a practice, and the course arc.
 - **Today:** what a Methods section has to tell you, and whether this one does.
 - **Thursday (L1):** build the merged NHANES frame those answers rest on.
 - **Toward M0 (Wk3):** ask the same six questions of two or three candidate papers.
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Slide 1 — Where we are in the project (3 min)

- Week 1 argued that a result you cannot rebuild is a claim, not a finding.
- Today is the step before rebuilding: knowing **what** you are trying to rebuild.
- By Thursday you need a merged NHANES table; by Week 3 you need a locked paper.

Slide 2 — Why the question comes before the data (4 min)

- The instinct is to start with “what did they run?”. A model family answers nothing on its own: a Cox model appears under four different research goals.
- Reading a Methods section is the first act of reproduction. You do it to Kueh today, then to your own candidates before Week 3.
- *MASLD hook:* by the end of today you will have watched this paper answer four of the six questions clearly, one partially, and one not at all.

Slide 3 — The six questions (6 min)

1. What is the **research goal**?
2. Who is the **population**, and how is eligibility operationalised?

3. What is the **exposure / index factor**, and the comparison?
 4. What is the **outcome**, time zero, and follow-up?
 5. What **other variables** enter the analysis, and what role are they supposed to play?
 6. What **design and analysis features** must be respected?
- Keep this slide visible: every section below fills one row.
 - **If any one is under-specified, the study is not reconstructable — no matter how clean the results look.**
 - These six are not this paper's; they are the template you take to your own at M0.

Q1 — What is the research goal? · 15 minutes across the slides below

Slide 4 — The paper states two aims, not one (4 min)

- Aim 1: the **clinical correlates** of four groups defined by BMI and waist-to-height ratio.
- Aim 2: the **prognostic value** of that classification for mortality.
- Related, but not the same question: the first is cross-sectional at baseline, the second needs follow-up. One paper can carry more than one research question.

Slide 5 — Five goals the same variables can serve (5 min)

Goal	What it asks
Description	What are the characteristics and mortality experience of each group?
Association	Is phenotype statistically associated with subsequent mortality?
Prognostic factor	Does phenotype inform mortality over and above other known factors?
Prediction	How accurately can we predict an individual's risk; does WHtR improve it?
Causal	What would mortality have been under a specified change in central adiposity?

- The variables can be identical. **The target quantity is not.**

Slide 6 — Different goals need different evidence (6 min)

- An **HR with a CI** is enough to report an adjusted association or a prognostic effect.
- It is **not** evidence that a prediction model is good — that needs calibration and discrimination — and not by itself evidence that a causal effect was identified.
- *MASLD hook*: the paper names “prognostic value” and supplies adjusted association. That gap is invisible unless you ask what evidence each goal would require.
- Vote and argue: which row is this paper actually on?

Q2 — Who is the population, and how is eligibility operationalised? · 16 minutes across the slides below

Slide 7 — NHANES in one picture (5 min)

- Continuous NHANES, released in **2-year cycles**; roughly **5,000 examined persons per cycle** after screening; six cycles stacked here (2007–2018).

- Three collection settings, and the distinction matters in a moment: **Interview** at home · **MEC exam** (measured body size, blood pressure) · **Laboratory** (triglycerides, GGT, glucose; some assays only in the fasting subsample).
- Data arrive as **many small topic files**, not one table.

Slide 8 — SEQN is the join key (4 min)

- Every respondent has one **SEQN**, the identifier CDC uses to sort and merge NHANES files.
- Analysis = pick the topic files you need, then **merge them on SEQN** into one row per person.
- SEQN does **not** repeat across the continuous cycles: all 59,842 values across 2007–2018 are distinct and the per-cycle ranges do not overlap. Joining on SEQN alone across stacked cycles is safe. *Analysis set: all NHANES 2007–2018 records, $N = 59,842$.*
- Keep a **cycle variable** anyway — provenance, definitions that change between cycles, and weights built per cycle. Not to disambiguate SEQN.
- *MASLD hook*: reproducing Table 1 means merging DEMO + BMX + BIOPRO + TRIGLY + ALQ + mortality, all keyed on SEQN across six cycles.

Slide 9 — “MASLD” is a derived variable, not a diagnosis (4 min)

- MASLD = hepatic steatosis **plus** at least one cardiometabolic criterion.
- Gold-standard steatosis is biopsy or imaging. **NHANES has neither** in this window.
- So the paper uses a surrogate: the **Fatty Liver Index**, thresholded at **FLI ≥ 60** , computed from **triglycerides, BMI, GGT and waist**.
- The disease definition forces the workaround; it is a feature of the data, not a shortcut.

Slide 10 — The funnel, with a type on every row (3 min)

Step	N	Kind of step
NHANES 2007–2018 records	59,842	—
Age ≥ 18	36,580	eligibility
In fasting subsample	14,962	design — drops 21,618
FLI computable	14,170	missingness
Steatosis (FLI ≥ 60)	6,052	target population
MASLD (≥ 1 criterion)	6,050	target population
Mortality-linked	6,048	outcome ascertainment

Analysis set: locked mortality-linked cohort, $N = 6,048$.

- A funnel shows N falling; the **type** column shows why. The largest exclusion is not clinical — it is a design feature, because the FLI needs fasting labs.
- Check that the prose arithmetic actually lands on the reported N. Ours does not match the paper’s ~6,300, and saying so is the work.

Slide 11 — BREAK (10 min)

Q3 — What is the exposure, and the comparison? · 15 minutes across the slides below

Slide 12 — The exposure is constructed, not measured (5 min)

- Obesity by **BMI** ≥ 30 (Asian participants ≥ 25) $\times$ central adiposity by **WHtR** ≥ 0.60 , giving four phenotype groups.
- Group IV = not obese by BMI, high waist-for-height — the cell the paper is about.
- *MASLD hook*: the exposure spans **two collection settings** — BMI and waist from the MEC exam — so a person missing either is missing the exposure.

Slide 13 — Which group is the comparison? (4 min)

- There is no treated and control arm. The four-category phenotype is the index factor, and mortality is compared **across** the categories.
- **Group I is the reference.** Every hazard ratio in the paper is measured against it.
- A ratio without its referent means nothing — and it is counterintuitive here that the referent is an obese group.

Slide 14 — Eligibility and exposure share their inputs (6 min)

Inputs	
Eligibility (FLI)	triglycerides, GGT, BMI , waist
Exposure (phenotype)	BMI , waist $\div$ height

- Being selected into the study and being placed in a phenotype group are **not independent events**. That is **structural dependence**.
- What it does to an estimate depends on the research goal and the causal structure — which is Q1's lesson arriving with consequences. We name it today; **M1** dissects it.
- Takeaway for any paper: **never trust a label you have not recomputed**.

Q4 — What is the outcome, time zero, and follow-up? · 11 minutes across the slides below

Slide 15 — Outcome and the clock (6 min)

- Outcome: **all-cause mortality** (MORTSTAT), via the NCHS linked mortality file — not an NHANES file, a separate linked, fixed-width one.
- **Time zero is the MEC examination.** PERMTH_EXM is person-months from that exam.
- Follow-up runs through **31 December 2019**, stated in the NCHS 2019 Linked Mortality File linkage methodology document — not inferred from the filename.
- Participants entered between 2007 and 2018, so follow-up lengths differ. That is why this is a survival analysis and not a risk difference.

Slide 16 — Time zero is a question the checklist never asks (5 min)

- “Exposure, outcome, covariates” can all be specified while time zero stays ambiguous, and an ambiguous time zero silently changes who is at risk and for how long.
- *MASLD hook*: exposure here is measured **at** time zero, which is convenient and rare — in your own papers, ask when the exposure was measured relative to the clock starting.
- Ask of your M0 candidates: when does follow-up begin, and does everyone start together?

Q5 — What other variables enter, and in what role? · 14 minutes across the slides below

Slide 17 — In the model is not the same as confounding (6 min)

- The paper adjusts for age, sex, and a list of comorbidities and medications.
- “We adjusted for X” says where X sits in the syntax. It does not say what X **does**.
- The same variable can be a confounder, a mediator, a competing prognostic factor, or a collider — and which it is depends on the research goal from Q1.

Slide 18 — What a crude-to-adjusted change establishes (8 min)

- Group IV all-cause HR $\approx$ **15.1 crude** $\rightarrow \approx$ **2.9 adjusted**. *Analysis set: full reconstructed analytic file, $N = 6,371$.*
 - Conditioning on the adjustment set moves the estimate enormously. What that movement **means** — confounding controlled, mediation removed, a collider opened — depends on the goal and on the causal role of those variables.
 - **Week 4 does that work**. Today it is enough to see that the size of a move is not itself an interpretation.
 - Vote and argue: what would you need to know to call this confounding?
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Q6 — What design and analysis features must be respected? · 11 minutes across the slides below

Slide 19 — NHANES is not a simple random sample (5 min)

- A complex multistage probability design: **weights, strata (SDMVSTRA), PSUs (SDMVPSU)**. We *use* them in Week 6; today, note they exist and that ignoring them is a choice with consequences.
- Fasting-subsample analyses carry their own weight, **WTSF2YR**, not **WTMEC2YR**.

Slide 20 — Designed subsampling is not missingness (6 min)

- The fasting subsample is **statistically designed**, not self-selected, and carries a weight built to keep it nationally representative.
 - Check it rather than assume: adults in and out of it differ by **0.2 years of age, 0.02 kg/m² of BMI and 0.07 cm of waist**. It is balanced on the very measurements this paper’s exposure is built from. *Analysis set: all NHANES adults 2007–2018, $N = 36,580$.*
 - So the distinction is **designed subsampling $\neq$ ordinary missingness** — not “the fasting subsample is a biased sample”.
 - *MASLD hook*: the paper’s Statistical analysis section never names the survey design. That is Q6 answered by omission, and it is the honest finding to report.
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Slide 21 — M0: ask the six of your own papers (12 min)

- Shortlist two or three candidate NHANES papers and answer all six questions for each, each answer with a pointer to where in the paper you found it.
- **Reconstructable is necessary, not sufficient**. A paper can be perfectly specified and still be a bad M0 choice — too few events, no design feature worth respecting, no variable whose handling you could interrogate.
- The M0 criteria ask for exactly this: enough observations and events, a design feature its analysis must respect, and a missing variable the analysis actually uses.

- Where a paper leaves a question unanswered, that is not a disqualification — it is a candidate for the judgement you will defend at M1.

Slide 22 — Bridge to today’s lab (L1) and M0 (5 min)

- **In L1 (Thu):** download several NHANES cycles, read the raw topic files, and merge them on `SEQN` into one analytic-ready table with a cycle indicator.
 - **How it becomes your project:** L1’s merge is the substrate for the whole term — Table 1, confounding, interaction, survey weights and missing data all run on it.
 - **Toward M0 (Wk3):** bring six answers for each candidate paper, not a summary.
 - Takeaway: reading a Methods section *is* the first act of reproduction.
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Deliberately deferred

Four things the old Week 2 taught are **not** in the rebuild, and are deferred on purpose rather than squeezed under a question:

- **The alcohol-exclusion three-way demonstration** (~23 min including live work). It is a cohort-construction exercise and Week 3 owns it (`Week3_cohort_construction_FLI.md`). It would have been the largest single block in a lecture whose job is a framework.
- **“Table 1 reproduces closely, therefore the variable definitions are recoverable.”** The inference is exactly what Q5 warns against, and Week 4 carries the careful version.
- **The reading-Methods traps checklist.** Superseded: the six questions are the checklist. One line survives at Slide 14 — never trust a label you have not recomputed.
- **The four primitives** (exposure / outcome / eligibility / variables). Kept as an idea, demoted from *the* framework to a reconstructability check nested inside Q2–Q5. It never asked what kind of question the variables were meant to answer, which is why a crude-to-adjusted collapse could be read as settled confounding.

If a later revision wants any of these back, they belong where they are now owned, not retrofitted into a six-question spine.